

Project Design Phase Solution Architecture

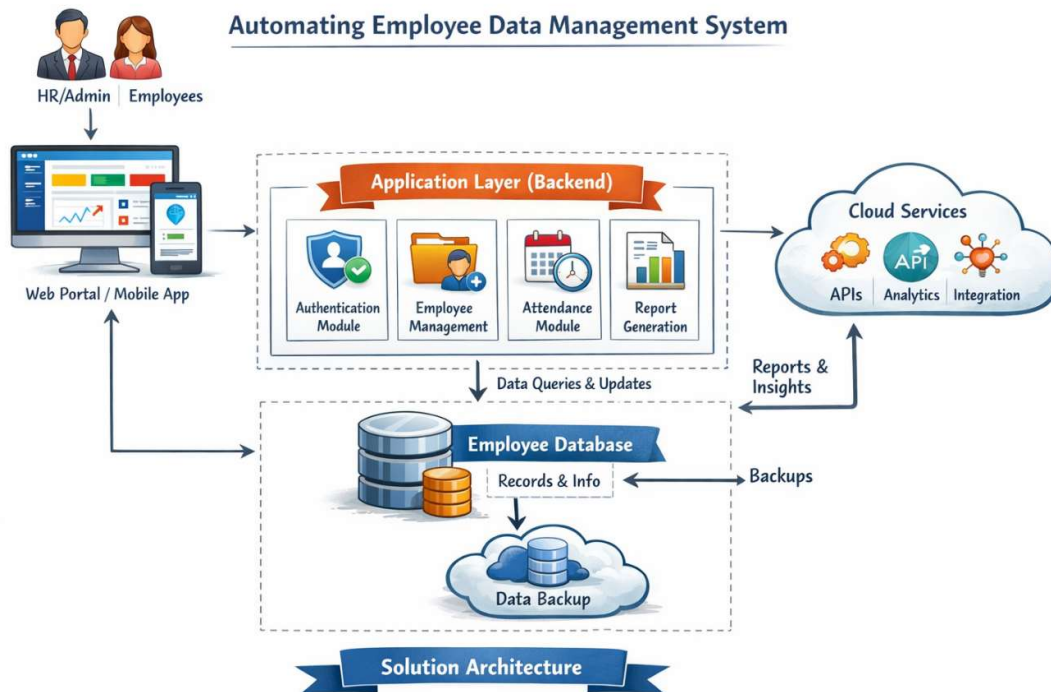
Date	05 April 2026
Team ID	SWTID-2026-4745
Project Name	Automating Employee Data
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is the process of designing a structured framework that connects business problems with appropriate technology solutions. It defines how different components of the system interact with each other and ensures that the system meets business requirements effectively.

The main goal of this solution architecture is to develop an efficient and scalable system for managing employee data in an automated way.

Example - Solution Architecture Diagram:



1. Overview of the Architecture

The proposed system follows a three-tier architecture:

- **Presentation Layer (Frontend):**
This is the user interface where HR and employees interact with the system. It can be developed using web technologies like HTML, CSS, and JavaScript.
- **Application Layer (Backend):**
This layer processes user requests, applies business logic, and controls the system functionalities such as data validation, authentication, and report generation.
- **Data Layer (Database):**
This layer stores all employee-related data such as personal details, attendance, and records securely in a database (e.g., MySQL).

2. System Components

- **User Interface:** Allows users to log in, view, and update employee data.
- **Authentication Module:** Ensures secure login using username and password.
- **Employee Management Module:** Handles adding, updating, and deleting employee records.
- **Attendance Module:** Tracks employee attendance automatically.
- **Report Generation Module:** Generates reports for management analysis.
- **Database:** Stores all system data securely.

3. Data Flow

1. The user logs into the system through the frontend.
2. The request is sent to the backend server.
3. The backend validates the request and processes it.
4. Data is retrieved or updated in the database.
5. The response is sent back to the user interface and displayed.

4. Key Features of the Architecture

- Secure and role-based access control
- Real-time data processing
- Scalable and modular design
- Easy integration with future technologies
- Reliable data storage and backup

5. Development Phases

1. Requirement Analysis
2. System Design
3. Development
4. Testing
5. Deployment
6. Maintenance

6. Conclusion

This solution architecture ensures that the employee data management system is efficient, secure, and scalable. It bridges the gap between business needs and technical implementation by clearly defining system structure and workflow.

